From Calibration to Closure in a Compton-Core Finite-Filament Model:
Spherical and NLS Pressure-Cell Closure for the Fine-Structure Scale

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Abstract

A previous Compton-core finite-filament scaling model showed that several electron-scale quantities can be fixed algebraically once a dimensionless core aspect scale is supplied. This note develops the pressure-cell closure used by the companion BEM/NLS calculation. The strongest derived element is the pressure-sector count $N_p = 4$, obtained from the spherical surface spectrum and supported by nonlinear and global topology checks. The pressure scale

$$E_p^{\mathrm{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right)$$

is therefore a conditional finite-cell closure: the factor $16\pi/3$ still requires the sector-volume normalization $4\pi/3$ per pressure sector per unit ropelength, and the coefficient $11/48$ requires the unweighted shell normalization $w_\perp = 1$. The latter is not yet derived from the full GP/NLS core dynamics; the supplied relaxed-core proxy returns $w_\perp \simeq 2.076$. For $\mathcal{L}_K = 16.371637$, the conditional scale is $E_p^{\mathrm{NLS}} = 274.0748756568$. This paper should therefore be read as a structured pressure-cell/NLS-shell closure and gate isolation, not as a stand-alone first-principles derivation of the electromagnetic fine-structure constant.

1 Purpose and status

The objective is to separate a numerical calibration from a mathematical closure. Let $\mathcal{E}_0$ denote a zeroth-order topological aspect scale and let $\mathcal{L}_K = L_K/D$ denote the ropelength of a diameter-D ideal trefoil centreline. The present paper proves the following conditional result:

> Given the finite-core Biot–Savart self-energy, the circulation relation $\Gamma_0 = 2\pi r_c\|\mathbf{v}_{\circlearrowleft}\|$, the spherical pressure-cell functional of Sec. 7, and the aspect-renormalization rule of Sec. 9, the numerical fine-structure scale is fixed by $\mathcal{E}_0$ and $\mathcal{L}_K$.

The proof is conditional because $\mathcal{E}_0$ itself is not derived here from primitive field equations. Nor is the far-field electromagnetic $1/r$ law derived. Those are promoted to explicit open theorems in Sec. 12. This keeps the paper falsifiable: the model succeeds as a closure theorem if the stated assumptions are accepted, and it becomes a fundamental derivation only if those assumptions are later obtained from a unique dynamical principle.

The uniqueness of this finite-cell reduction is treated in the standalone companion Paper IV. In the present paper the same assumptions are used as the pressure-cell closure; Paper IV proves that, within the stated minimal $SO(3)$-covariant variational class, these assumptions uniquely select $N_p = 4$, $\sigma = 11/3$, and $\chi_R = 2$.

2 Derived-label gate and assumption ledger

The word "derived" is reserved here for quantities obtained from a stated variational problem, boundary-value problem, or theorem without using the empirical fine-structure constant, CODATA, m_e, e, R_∞, or $\hbar$ as numerical inputs. Under this convention the local logarithmic coefficient $A_K \to 1/(4\pi)$ is derived, but the pressure-cell coefficients introduced below are not yet first-principles derived. They are structured closure hypotheses.

Table 1: Status of the pressure-cell ingredients in this paper.

Item	Expression	Current status	Requirement for "derived" l
Local logarithmic coefficient	$A_K \to 1/(4\pi)$	Theorem	Already derived from the lo
Zeroth pressure prefactor	$16\pi/3$	Minimal-shell closure hypothesis	Derive or independently just
Finite-shell correction	$11/(48\mathcal{L}_K^2)$	Minimal-shell closure hypothesis	Derive the unweighted $3+2$
Cell radius closure	$R_{\mathrm{cell}} = 2L_K$	Minimal-shell closure hypothesis	Derive the self-dual inner/ou

Thus this paper supplies a coherent analytical closure framework. It does not by itself prove that the electromagnetic fine-structure constant is derived from first principles. That stronger label requires deriving the entries marked as closure hypotheses in Table 1.

2.1 Why the minimal pressure shell has four sectors

The pressure manifold is taken to be the lowest $SO(3)$-invariant boundary subspace capable of representing both uniform compression and displacement of the cell pressure centre. The pressure field on a spherical cell boundary admits the harmonic expansion

$$p(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} p_{\ell m} Y_{\ell m}(\theta, \phi).$$

Uniform compression is the monopole sector $\ell = 0$, with degeneracy $2\ell + 1 = 1$. The lowest pressure-gradient/displacement sector is $\ell = 1$, with degeneracy 3. Therefore the minimal pressure manifold is

$$\mathcal{P}_{\min} = \mathcal{H}_0 \oplus \mathcal{H}_1, \qquad N_p = \dim \mathcal{H}_0 + \dim \mathcal{H}_1 = 1 + 3 = 4.$$

Modes $\ell \geq 2$ are quadrupolar and higher shape deformations. In a surface-stiffness action containing $\int_{S^2} |\nabla_{S^2} p|^2 d\Omega$, they carry eigenvalues $\ell(\ell+1)$ and enter as higher-order shape corrections, not as zeroth-order pressure sectors. The detailed coefficient gates are summarized in Appendix C. For a complete robustness claim, the companion numerical package should also report an $\ell \leq 2$ extension as a negative or sensitivity control. If the quadrupolar sector shifts the result at leading order, the $\ell \leq 1$ cutoff must be treated as a fit-relevant closure rather than a derived low-mode truncation.

2.2 Why the shell deficit is $3 + \langle \sin^2 \theta \rangle$

The finite-shell deficit is decomposed into a spherical volume second variation and a transverse NLS-shell contribution,

$$\sigma = \sigma_{\mathrm{vol}} + \sigma_{\perp}.$$

For a relative inward shell displacement $\eta = a/R_{\mathrm{cell}}$, the spherical volume Jacobian gives

$$\frac{V(R-a)}{V(R)} = (1-\eta)^3 = 1 - 3\eta + 3\eta^2 + O(\eta^3).$$

After the first-order displacement has been absorbed into the equilibrium radius, the second-order volume coefficient is

$$\sigma_{\text{vol}} = 3.$$

The isotropic NLS/GP gradient term contributes through the transverse component of the shell deformation. For an isotropic shell orientation,

$$\sigma_\perp = \langle \sin^2\theta \rangle_{S^2} = \frac{\int_0^\pi \sin^2\theta \, \sin\theta \, d\theta}{\int_0^\pi \sin\theta \, d\theta} = \frac{2}{3}.$$

Thus

$$\sigma = 3 + \frac{2}{3} = \frac{11}{3}.$$

More generally one may write

$$\sigma = 3 + \frac{2}{3} w_\perp,$$

where $w_\perp = 1$ is the isotropic NLS-gradient weighting used by the minimal model. This parameterization makes the remaining vulnerability explicit: a first-principles derivation must obtain $w_\perp = 1$, not tune it to the empirical fine-structure constant. With $\eta_K = 1/(2\chi_R \mathcal{L}_K)$ and $\chi_R = 2$, this gives

$$\sigma \eta_K^2 = \frac{11}{3} \frac{1}{16\mathcal{L}_K^2} = \frac{11}{48\mathcal{L}_K^2}.$$

2.3 Why $A_\chi = \chi_R + N_p/\chi_R$ is the minimal radius action

Let $\chi_R = R_{\text{cell}}/L_K$. The minimal inner/outer pressure balance has two reciprocal costs. The outer pressure-volume cost grows with the dimensionless cell radius,

$$A_{\text{out}} \propto \chi_R,$$

whereas the inner crowding/compliance cost decreases with radius and is proportional to the number of pressure sectors,

$$A_{\text{in}} \propto \frac{N_p}{\chi_R}.$$

With self-dual inner/outer normalization, the lowest-order reciprocal action is

$$A_\chi(\chi_R) = \chi_R + \frac{N_p}{\chi_R}.$$

Stationarity gives

$$\frac{dA_\chi}{d\chi_R} = 1 - \frac{N_p}{\chi_R^2} = 0, \qquad \chi_R = \sqrt{N_p} = 2.$$

A more general action $a\chi_R + bN_p/\chi_R$ would give $\chi_R = \sqrt{bN_p/a}$. Hence $\chi_R = 2$ is derived within the self-dual inner/outer pressure normalization $a = b$. Without that self-duality assumption, $\chi_R = 2$ remains a sensitivity parameter.

2.4 Post-review gate-script updates

Three gate calculations update the status of the pressure-cell coefficients.

First, the pressure-sector count is now supported by the fixed-volume surface spectrum of a spherical cell. For a perturbation $r(\Omega) = R[1 + \epsilon f(\Omega)]$, the fixed-volume surface variation gives

$$\delta^2 A = \frac{R^2}{2} \int_{S^2} \left(|\nabla_{S^2} f|^2 - 2f^2 \right) d\Omega.$$

For $f = Y_{\ell m}$,

$$k_\ell = \ell(\ell+1) - 2 = (\ell-1)(\ell+2).$$

Thus $\ell = 1$ is the translation/centre-displacement zero-mode sector, $\ell \geq 2$ are positive-stiffness shape modes, and $\ell = 0$ is the scalar compression sector. Hence

$$N_p = \dim \mathcal{H}_0 + \dim \mathcal{H}_1 = 1 + 3 = 4$$

is derived in the spherical finite-cell surface reduction.

This cutoff is also robust beyond the quadratic spectrum. A perturbative finite-core audit verifies that $\ell \geq 2$ remains separated from the retained sector when $\|\delta H\|_{\rm op} < k_2/2 = 2$, and the physical scale $\eta_K = 1/(4\mathcal{L}_K) = 1.52703116982 \times 10^{-2}$ is far below that bound. A nonlinear star-shaped solve through $\ell_{\max} = 6$ tested 45 real $\ell \geq 2$ modes. Across 300 random samples per amplitude and 20 local minimizations from amplitude-one random starts, no competing leading $\ell \geq 2$ pressure mode was found. Global finite-perimeter cells are protected by the isoperimetric inequality $P(\Omega)^3 \geq 36\pi V(\Omega)^2$, with equality only for a ball, so non-star-shaped, disconnected, or higher-genus admissible cells cannot lower the fixed-volume area below the sphere.

Second, the GP core-profile second-variation audit solves the radial GP/NLS vortex profile and confirms that the angular transverse factor is profile independent under isotropic radial weights:

$$\left\langle \sin^2\theta \right\rangle_{S^2} = \frac{2}{3}.$$

The coefficient is therefore

$$\sigma = 3 + \frac{2}{3} w_\perp.$$

For the unweighted isotropic shell normalization $w_\perp = 1$, one obtains

$$\sigma = \frac{11}{3}, \qquad \frac{\sigma}{4\chi_R^2} = \frac{11}{48} \quad (\chi_R = 2).$$

Sensitivity runs with alternative profile weights do not generically reproduce 11/48. Hence the coefficient is derived within the unweighted isotropic GP/NLS shell reduction, and the remaining microscopic gate is to derive $w_\perp = 1$ from the full primitive reduction.

Third, the reciprocal radius action is now tied to the SST pressure identity. With the canonical constants,

$$\frac{1}{2}\rho_{\rm core}\|\mathbf{v}_{\circlearrowleft}\|^2 = \frac{F_{\rm swirl}^{\max}}{2\pi r_c^2} = 2.329244600448 \times 10^{30}\ \text{Pa}.$$

Thus $P_{\rm in}/P_{\rm out} = 1$ in the leading Laplace-matched reciprocal model, giving

$$A_\chi = \chi_R + \frac{N_p}{\chi_R}, \qquad \chi_R = \sqrt{N_p} = 2.$$

The remaining robustness requirement is to exclude same-order nonreciprocal terms such as $\log \chi_R$, χ_R^2, or χ_R^{-2}.

3 Claim-status correction: $16\pi/3$ and $11/48$

The surface-spectrum calculation derives $N_p = 4$, but this does not by itself derive the numerical prefactor $16\pi/3$. The additional step is

$$E_p^{(0)} = N_p \left(\frac{4\pi}{3}\right) \mathcal{L}_K,$$

which assumes a sector-volume normalization $4\pi/3$ per pressure sector per unit ropelength. This is a compact and useful closure ansatz, but it is still a load-bearing open coefficient until obtained from a variational pressure-cell principle.

Similarly,

$$\sigma = 3 + \frac{2}{3}w_{\perp}$$

is the derived shell structure. The value $11/48$ follows only after setting $w_{\perp} = 1$ and $\chi_R = 2$. The current relaxed-core proxy gives $w_{\perp} \simeq 2.076$, so the primitive GP/NLS derivation of $w_{\perp} = 1$ is not established. All precision claims that use $11/48$ are therefore conditional on the unweighted isotropic shell normalization.

4 Finite-core filament energy

Consider a smooth closed filament centreline $X(s) \subset \mathbb{R}^3$, parametrized by arclength $s \in [0, L_K]$, with unit tangent $T(s) = \partial_s X(s)$. Let a be a small core cutoff satisfying

$$a \ll \ell_{\text{curv}}, \qquad a \ll d_{\min},$$

where ℓ_{curv} is the local curvature scale and $d_{\min}$ is the minimum nonlocal self-distance of the centreline.

The dimensionless cutoff Biot–Savart energy used in the numerical implementation is

$$E_{\text{BS}}^{\text{norm}}(a) = \frac{1}{8\pi}\int_0^{L_K}\int_0^{L_K} \frac{T(s)\cdot T(s')}{\|X(s)-X(s')\|}\,\Theta\big(\|X(s)-X(s')\|-a\big)\;ds\,ds', \tag{1}$$

where Θ is a cutoff indicator. Its physical normalization is

$$E_{\text{BS}}(a) = \rho_f \Gamma_0^2 E_{\text{BS}}^{\text{norm}}(a). \tag{2}$$

For a slender filament, the fitted local form is

$$E_{\text{BS}}(a) = \rho_f \Gamma_0^2 L_K \left[A_K \log\left(\frac{L_K}{a}\right) + a_K\right] + o(1) \quad (a/L_K \to 0). \tag{3}$$

5 Local logarithmic coefficient

Theorem 1 (Universal local Biot–Savart logarithm)**.** *For any C^2 smooth slender closed filament, the leading local logarithmic coefficient in Eq. (3) is*

$$A_K = \frac{1}{4\pi}$$

provided the energy is normalized as in Eq. (1).

Proof. Fix s and write $u = s' - s$. In the near-diagonal region $a < |u| < \ell$, where $a \ll \ell \ll \ell_{\text{curv}}$, smoothness gives

$$X(s+u) - X(s) = uT(s) + O(u^2), \qquad T(s)\cdot T(s+u) = 1 + O(u^2).$$

Therefore

$$\frac{T(s)\cdot T(s+u)}{\|X(s+u)-X(s)\|} = \frac{1}{|u|} + O(1).$$

The local singular part of Eq. (1) per unit arclength is then

$$\frac{1}{8\pi}\int_{a<|u|<\ell} \frac{du}{|u|} = \frac{1}{8\pi}\left[2\log\left(\frac{\ell}{a}\right)\right] = \frac{1}{4\pi}\log\left(\frac{\ell}{a}\right).$$

Nonlocal contributions and curvature-dependent finite terms alter a_K, not the coefficient of $\log(1/a)$. Hence $A_K = 1/(4\pi)$. □

Remark 1. *This theorem is local. It does not assert that every finite-resolution numerical plateau is exactly* $1/(4\pi)$*; it asserts the slender-core asymptotic coefficient. Knot geometry enters the finite part* a_K*, the ropelength* $\mathcal{L}_K$*, and nonlocal corrections.*

6 Pressure balance and core closure

The normal pressure exerted by the exterior irrotational self-energy on a tube of radius a is the negative derivative with respect to volume. Since the tube area is $2\pi a L_K$,

$$P_{\mathrm{irr}}(a) = -\frac{1}{2\pi a L_K}\frac{\partial E_{\mathrm{BS}}}{\partial a}. \tag{4}$$

Using Eq. (3),

$$P_{\mathrm{irr}}(a) = \frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2}. \tag{5}$$

The core kinetic pressure scale is

$$P_{\mathrm{core}} = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2. \tag{6}$$

The no-contact pressure balance $P_{\mathrm{irr}} = P_{\mathrm{core}}$ gives

$$\frac{\rho_f \Gamma_0^2 A_K}{2\pi a^2} = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2, \tag{7}$$

hence

$$a = \frac{\Gamma_0}{\|\mathbf{v}_{\circlearrowleft}\|}\sqrt{\frac{A_K}{\pi}}. \tag{8}$$

With the circulation relation

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\|, \tag{9}$$

this becomes

$$\boxed{\frac{a_{\mathrm{nc}}}{r_c} = \sqrt{4\pi A_K}.} \tag{10}$$

Therefore, in the theorem limit $A_K = 1/(4\pi)$,

$$\boxed{a_{\mathrm{nc}} = r_c.} \tag{11}$$

Dimensional consistency is immediate:

$$[P_{\mathrm{irr}}] = \frac{[\rho_f][\Gamma_0]^2}{[a]^2} = \frac{\mathrm{kg\,m^{-3}\ m^4\,s^{-2}}}{\mathrm{m}^2} = \mathrm{kg\,m^{-1}\,s^{-2}} = \mathrm{Pa}.$$

7 Spherical pressure-cell functional

The local calculation fixes the tube radius. The next question is whether the exterior spherical cell can generate the finite correction required for the fine-structure scale.

Let a trefoil tube of total length L_K and diameter $D = 2a$ be embedded in a spherical coarse-graining cell of radius

$$R_{\mathrm{cell}} = 2L_K. \tag{12}$$

Then

$$\eta_K = \frac{a}{R_{\mathrm{cell}}} = \frac{a}{2L_K} = \frac{1}{4(L_K/D)} = \frac{1}{4\mathcal{L}_K}. \tag{13}$$

7.1 Geometric pressure-transfer factor

A small outward radial displacement of the spherical control surface by a gives the surface-area Jacobian

$$J_A(\eta_K) = \frac{4\pi(R_{\text{cell}} + a)^2}{4\pi R_{\text{cell}}^2} = (1 + \eta_K)^2 = 1 + 2\eta_K + \eta_K^2. \tag{14}$$

The isotropic radial volume-response mode contributes one additional linear factor,

$$J_V^{(r)}(\eta_K) = \eta_K. \tag{15}$$

The total spherical pressure-transfer multiplier is therefore

$$\boxed{\Xi_{\text{sph}} = J_A + J_V^{(r)} = 1 + 3\eta_K + \eta_K^2 = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.} \tag{16}$$

7.2 Explicit variational form

To make the above statement variational rather than merely kinematic, introduce an isotropic cell-response coordinate q. Let $K_{\text{sph}} > 0$ be a spherical compression modulus and define

$$\mathcal{G}_{\text{sph}}(q; \eta_K) = \frac{K_{\text{sph}}}{2} q^2 - K_{\text{sph}} \left[J_A(\eta_K) + J_V^{(r)}(\eta_K) \right] q. \tag{17}$$

Stationarity gives

$$\frac{\partial \mathcal{G}_{\text{sph}}}{\partial q} = K_{\text{sph}} \left[q - \left(J_A + J_V^{(r)} \right) \right] = 0, \tag{18}$$

hence

$$q_\star = \Xi_{\text{sph}}. \tag{19}$$

Since

$$\frac{\partial^2 \mathcal{G}_{\text{sph}}}{\partial q^2} = K_{\text{sph}} > 0, \tag{20}$$

the stationary point is a strict minimum.

Remark 2. *Equation (17) is an explicit coarse-grained pressure-cell functional. It does not yet derive $R_{\text{cell}} = 2L_K$ from primitive dynamics; Eq. (12) is a closure choice. A fully fundamental derivation must replace Eq. (12) by a stationarity condition on R_{cell}.*

The derivation-target lemmas for the pressure prefactor, the finite-shell coefficient, and the radius closure are collected in Appendix C. This appendix is part of the derived-label gate: it identifies the exact mode-count and shell-variation results still required to upgrade the pressure-cell scale from structured closure to first-principles derivation.

8 NLS finite-shell pressure scale

The spherical response factor in Sec. 7 fixes the finite-core transfer geometry, but the companion finite-cell eigenvalue problem also requires a pressure equilibrium scale and a stiffness. The non-linear Schrödinger/Gross–Pitaevskii vortex-core asymptotics provide a natural source for both quantities. For a large vortex ring with core radius r_c, the standard asymptotic form is

$$E_{\text{ring}}(R) = \frac{1}{2} \rho \Gamma^2 R \left[\ln\left(\frac{8R}{r_c} \right) - A \right], \tag{21}$$

and the translational velocity has the form

$$V_{\text{ring}}(R) = \frac{\Gamma}{4\pi R} \left[\ln\left(\frac{8R}{r_c} \right) - B \right]. \tag{22}$$

Hamiltonian consistency $V_{\rm ring} = dE_{\rm ring}/dP$, with $P(R) = \rho\Gamma\pi R^2$, gives

$$B = A - 1. \tag{23}$$

The Golden NLS branch uses

$$A = \varphi, \qquad B = \varphi^{-1}, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{24}$$

The zeroth-order spherical pressure scale associated with an ideal trefoil cell is

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K. \tag{25}$$

Finite tube thickness contributes a second-order shell correction in the small parameter

$$\eta_K = \frac{1}{4\mathcal{L}_K}. \tag{26}$$

The NLS finite-shell pressure scale used in the companion BEM calculation is

$$\boxed{E_p^{\rm NLS} = \frac{16\pi}{3}\mathcal{L}_K\left[1 - \frac{11}{48\mathcal{L}_K^2}\right] = \frac{16\pi}{3}\mathcal{L}_K\left[1 - \frac{11}{3}\eta_K^2\right].} \tag{27}$$

The corresponding shell stiffness is

$$\boxed{\kappa_{\rm NLS}^{\rm shell} = 8\pi\mathcal{L}_K^2 = \frac{\pi}{2\eta_K^2}.} \tag{28}$$

The coefficient 11/48 is not fitted to α, but it is conditional. The post-review GP core-profile audit derives the decomposition $\sigma = 3 + (2/3)w_\perp$. The unweighted isotropic shell normalization $w_\perp = 1$ gives $\sigma = 11/3$ and therefore $11/(48\mathcal{L}_K^2)$ at $\chi_R = 2$. However, the current relaxed-core proxy gives $w_\perp \simeq 2.076$. Thus a full primitive GP/SST core-profile reduction must still explain why the shell normalization relevant to the finite-cell pressure response is $w_\perp = 1$, or else the 11/48 correction must be replaced. The stiffness $\kappa_{\rm NLS}^{\rm shell}$ is the corresponding thin-shell stiffness; it is large because $\eta_K \ll 1$.

The finite-cell pressure enthalpy used by the companion BEM eigenvalue calculation is

$$\mathcal{A}_{\rm pressure}(E) = \kappa_{\rm NLS}^{\rm shell}\left[\frac{1}{3}\left(\frac{E}{E_p^{\rm NLS}}\right)^3 - \ln\left(\frac{E}{E_p^{\rm NLS}}\right)\right]. \tag{29}$$

It satisfies

$$\frac{d\mathcal{A}_{\rm pressure}}{d\log E} = \kappa_{\rm NLS}^{\rm shell}\left[\left(\frac{E}{E_p^{\rm NLS}}\right)^3 - 1\right], \tag{30}$$

so the pressure part alone has a strict stationary point at $E = E_p^{\rm NLS}$. The BEM part in the companion paper supplies the finite-cell compliance correction:

$$\mathcal{A}_{\rm total}(E) = \mathcal{A}_{\rm BEM}(E) + \mathcal{A}_{\rm pressure}(E), \qquad \frac{d\mathcal{A}_{\rm total}}{d\log E} = 0. \tag{31}$$

Thus the present paper supplies the analytical NLS-shell closure, while the companion calculation determines the stationary eigenvalue $E_\star$.

For the ideal trefoil value $\mathcal{L}_K = 16.371637$, Eq. (27) gives

$$E_p^{(0)} = 274.309410808, \tag{32}$$

$$E_p^{\rm NLS} = 274.074875657, \tag{33}$$

$$\kappa_{\rm NLS}^{\rm shell} = 6736.341149. \tag{34}$$

These are dimensionless geometric/NLS closure quantities; no electron mass, Planck constant, charge, Rydberg constant, or CODATA value is used to evaluate them. Their use as first-principles inputs remains conditional on deriving the prefactor $16\pi/3$, the shell coefficient 11/48, and the cell-radius closure.

9 Aspect renormalization and fine-structure scale

The pressure-cell closure assumes that the zeroth-order aspect scale $\mathcal{E}_0$ receives a small finite-core correction proportional to the local core coefficient and to the spherical transfer response. In the theorem limit $A_K = 1/(4\pi)$,

$$\pi^2 A_K \to \frac{\pi}{4}. \tag{35}$$

The effective aspect scale is therefore

$$\boxed{\mathcal{E}_{\text{eff}} = \mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2}\Xi_{\text{sph}}\right].} \tag{36}$$

The closure identification is

$$\boxed{\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_{\text{eff}}}.} \tag{37}$$

Combining Eqs. (16), (36), and (37) gives

$$\boxed{\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_0 \left[1 - \frac{\pi}{4\mathcal{E}_0^2}\left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right]}.} \tag{38}$$

Remark 3. *Equation (38) is a theorem inside the specified closure model. It is not yet an unconditional derivation of the QED coupling because the far-field interaction law has not been derived. The necessary additional theorem is stated in Sec. 12.*

10 Numerical evaluation

Use the ideal-trefoil ropelength value

$$\mathcal{L}_K = 16.371637, \tag{39}$$

taken from the ideal-knot Fourier data source. Then

$$\eta_K = \frac{1}{4\mathcal{L}_K} = 0.01527031170, \tag{40}$$

$$\Xi_{\text{sph}} = 1 + 3\eta_K + \eta_K^2 = 1.046044118, \tag{41}$$

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K = 274.309410808, \tag{42}$$

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right) = 274.074875657, \tag{43}$$

$$\kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2 = 6736.341149. \tag{44}$$

The companion BEM/NLS calculation uses E_p^{NLS} and $\kappa_{\text{NLS}}^{\text{shell}}$ in Eq. (31) and finds the stationary aspect eigenvalue $E_\star$. Once that eigenvalue is supplied, the finite-core effective aspect is evaluated as

$$E_{\text{eff}} = E_\star \left[1 - \frac{\pi}{4E_\star^2}\left(1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}\right)\right], \tag{45}$$

and the dimensionless fine-structure-scale estimate is

$$\alpha_{\text{cell}} = \frac{2}{E_{\text{eff}}}. \tag{46}$$

Table 2: Analytical pressure-cell quantities supplied to the companion BEM/NLS finite-cell eigenvalue calculation.

Quantity	Expression	Value
Ideal trefoil ropelength	$\mathcal{L}_K$	16.371637
Shell parameter	$\eta_K = 1/(4\mathcal{L}_K)$	0.01527031170
Spherical transfer	$\Xi_{\rm sph}$	1.046044118
Zeroth pressure scale	$E_p^{(0)} = (16\pi/3)\mathcal{L}_K$	274.309410808
NLS pressure scale	$E_p^{\rm NLS}$	274.074875657
NLS shell stiffness	$\kappa_{\rm NLS}^{\rm shell} = 8\pi\mathcal{L}_K^2$	6736.341149

10.1 Phase–pressure normal-mode closure of the far-field Hessian

The far-field stiffness gate has also been sharpened by the companion normal-mode audit. The exterior Hodge calculation gives

$$q_\phi = 1, \qquad \Lambda_\phi = 4\pi R\mathcal{K}_{\rm cell}.$$

The interior chain now proceeds as follows. The radial phase problem gives the optimal profile $n_{\rm eff}(r) \propto r^{-2}$. The accessible-area projection gives

$$M_{\rm acc} = M_\phi \frac{(R-a)^2}{R^2}.$$

Finally, in the canonical linearized Madelung/GP normal-mode reduction,

$$H_k = \frac{1}{2}p_k^2 + \frac{1}{2}\omega_k^2 q_k^2$$

has equal cycle-averaged phase/flow and pressure/compression budgets:

$$M_\phi = M_p = \frac{E_{\rm eff}}{2}.$$

Therefore

$$\Lambda_\phi = \frac{E_{\rm eff}}{2R^2},$$

and for $R = 1$,

$$\Lambda_\phi = \frac{E_{\rm eff}}{2}.$$

This closes the far-field Hessian gate within the canonical normal-mode reduction. The remaining bridge assumption is that the $E_{\rm eff}$ used in the pressure-cell chain is the cycle-averaged quadratic energy of that same interior canonical mode family.

11 Reproducibility files for the coefficient gates

The gate calculations referenced in this paper are stored in `../code/` relative to the `Manuscripts` directory:

- $N_p = 4$: `derive_pressure_mode_cutoff_surface_spectrum.py`, `test_finite_core_nonspherical_shap` and `solve_nonlinear_shape_stability.py`;
- $11/48$ conditional on $w_\perp = 1$: `derive_gp_core_profile_second_variation.py`;
- $w_\perp$ warning check: `derive_relaxed_gp_core_hessian_wperp.py`;

- $\chi_R = 2$: `derive_pressure_self_duality_from_laplace_matching.py`;
- $q_\phi = 1$ and exterior capacity: `solve_one_cell_hodge_phase_hessian.py`;
- $\Lambda_\phi = E_{\text{eff}}/2$ within the canonical phase–pressure normal-mode reduction: `derive_phase_pressure_excha`

The corresponding CSV and Markdown outputs are in the matching `../code/outputs_*` directories.

12 What remains for a fundamental derivation

A further independent selection principle is required for the trefoil itself. The present paper uses $K = 3_1$ because that is the candidate finite-core filament assigned to the electron sector in the model. This assignment is not derived here. The falsifiable requirement is to run the same closure for the unknot, the figure-eight knot 4_1, and other low-complexity knots and to show that they do not reproduce the observed fine-structure scale under the same coefficient rules.

The preceding sections provide a conditional mathematical closure. To upgrade it to a fundamental derivation, the following results must be obtained without using α, R_∞, e, or CODATA as input.

12.1 Updated open gates after the post-review scripts

The current open gates are narrower than in the original closure statement. The pressure-sector count $N_p = 4$ is derived in the spherical surface variation, and $\chi_R = 2$ is derived in the Laplace-matched reciprocal pressure model. What remains is:

(G1) derive the unweighted transverse normalization $w_\perp = 1$ from the full GP/SST core-profile reduction;

(G2) extend the nonlinear shape-stability audit beyond star-shaped finite-perimeter cells only if the physical pressure-cell admissibility class is enlarged beyond reduced boundaries;

(G3) identify the E_{eff} entering the pressure-cell chain with the cycle-averaged quadratic energy of the same canonical interior mode family used in the phase–pressure normal-mode reduction.

12.2 Open Theorem I: topological aspect eigenvalue

The number

$$\mathcal{E}_0 = 274.074996 \tag{47}$$

must be obtained as an eigenvalue of a well-posed geometric or hydrodynamic variational problem:

$$\delta\mathcal{F}[X, \mathbf{u}, p, R_{\text{cell}}, a] = 0, \qquad \nabla \cdot \mathbf{u} = 0, \tag{48}$$

with trefoil topology and circulation constraints. Without this theorem, $\mathcal{E}_0$ remains an external aspect input.

12.3 Open Theorem II: spherical radius from stationarity

The closure radius $R_{\text{cell}} = 2L_K$ should be replaced by

$$\frac{\partial \mathcal{F}}{\partial R_{\text{cell}}} = 0. \tag{49}$$

If this stationarity condition yields $R_{\text{cell}} = 2L_K$, then $\eta_K = 1/(4\mathcal{L}_K)$ becomes dynamical rather than chosen.

12.4 Open Theorem III: far-field interaction law

A fine-structure constant is not only a number. In conventional electrodynamics it is the coefficient of a long-range coupling. The required far-field theorem is

$$V_{\text{eff}}(r) = \frac{\alpha_{\text{cell}}\hbar c}{r} + O(r^{-2}) \qquad (r \to \infty). \tag{50}$$

Equivalently, the model must yield

$$\alpha_{\text{cell}} = \frac{g_{\text{eff}}^2}{4\pi\hbar c}. \tag{51}$$

Only after Eq. (50) is derived from the same variational structure can Eq. (38) be called a fundamental derivation of the electromagnetic coupling.

13 Falsification criteria

The closure chain is falsified, or at least not fundamental, if any of the following fail:

(F1) The slender-core coefficient does not converge to $A_K = 1/(4\pi)$.

(F2) The ideal-trefoil ropelength used for $\mathcal{L}_K$ is not the relevant minimizing geometry.

(F3) A pressure-cell variational problem does not yield $R_{\text{cell}} = 2L_K$ or an equivalent $\eta_K = 1/(4\mathcal{L}_K)$.

(F4) The aspect-renormalization law Eq. (36) cannot be derived from a more primitive pressure or action principle.

(F5) The far-field interaction is not of the $1/r$ form in Eq. (50).

14 Conclusion

The standalone uniqueness proof in Paper IV shows that the pressure manifold, shell coefficient, and radius closure are not independent tunings once the minimal finite-cell variational reduction is imposed. Without Paper IV they should be read as closure hypotheses; with Paper IV they are consequences of that stated reduction.

The pressure-coefficient gates are stated explicitly in Appendix C; until those gates are closed, the NLS pressure scale should be called closure-derived rather than first-principles derived.

The mathematical route from calibration to pressure-cell closure is now separated into three levels. First, the local Biot–Savart singularity gives the slender-core coefficient $A_K \to 1/(4\pi)$, and the pressure balance then fixes $a_{\text{nc}} = r_c$. Second, the spherical control-volume response gives

$$\Xi_{\text{sph}} = 1 + \frac{3}{4\mathcal{L}_K} + \frac{1}{16\mathcal{L}_K^2}.$$

Third, the NLS finite-shell closure supplies the pressure scale and stiffness

$$E_p^{\text{NLS}} = \frac{16\pi}{3}\mathcal{L}_K\left(1 - \frac{11}{48\mathcal{L}_K^2}\right), \qquad \kappa_{\text{NLS}}^{\text{shell}} = 8\pi\mathcal{L}_K^2.$$

For the ideal trefoil these evaluate to

$$E_p^{\text{NLS}} = 274.074875657, \qquad \kappa_{\text{NLS}}^{\text{shell}} = 6736.341149.$$

These quantities are computed from the ropelength geometry and the stated NLS-shell closure hypotheses, not from α, m_e, $\hbar$, e, or R_∞. They should therefore be labelled as closure-derived rather than first-principles derived until the coefficients in Table 1 are obtained from an independent variational calculation. The stationary aspect eigenvalue $E_\star$ is obtained in the companion BEM/NLS finite-cell calculation. The final identification with the electromagnetic fine-structure constant further requires the far-field interaction theorem

$$V(r) = \frac{\alpha_{\text{cell}}\hbar c}{r} + O(r^{-2}).$$

A Dimensional checks

The three key energy terms have dimensions

$$[\rho_f \Gamma_0^2 L_K] = \mathrm{kg\,m^{-3}\,m^4\,s^{-2}\,m} = \mathrm{J}, \tag{52}$$

$$\left[\frac{1}{2}\pi\rho_f \|\mathbf{v}_\circlearrowleft\|^2 a^2 L_K\right] = \mathrm{kg\,m^{-3}\,m^2\,s^{-2}\,m^2\,m} = \mathrm{J}, \tag{53}$$

$$[P_{\text{irr}}] = \mathrm{Pa}. \tag{54}$$

The quantities A_K, $\mathcal{L}_K$, η_K, Ξ_{sph}, $\mathcal{E}_0$, $\mathcal{E}_{\text{eff}}$, and α_{cell} are dimensionless.

B Small-parameter expansion

Let

$$\epsilon = \frac{1}{\mathcal{E}_0}.$$

Then Eq. (36) is

$$\mathcal{E}_{\text{eff}} = \mathcal{E}_0\left[1 - \frac{\pi}{4}\Xi_{\text{sph}}\epsilon^2\right].$$

Therefore

$$\alpha_{\text{cell}} = \frac{2}{\mathcal{E}_0}\left[1 + \frac{\pi}{4}\Xi_{\text{sph}}\epsilon^2 + O(\epsilon^4)\right].$$

Since $\epsilon \simeq 3.65\times10^{-3}$, the correction is naturally of order 10^{-5}, matching the observed difference between the zeroth-order $2/\mathcal{E}_0$ estimate and the refined closure.

C Derivation-target lemmas for pressure-cell coefficients

This appendix records the exact gates that must be closed before the pressure-cell scale can be labelled first-principles derived. The statements below are intentionally split into algebraic implications, which are proved, and physical inputs, which remain derivation targets.

C.1 Lemma A1: spherical pressure-volume factor

Let $\mathcal{L}_K = L_K/D$ be the ideal-trefoil ropelength. Suppose the zeroth-order pressure-cell action decomposes into N_p equivalent spherical pressure sectors, each contributing the normalized spherical volume factor $4\pi/3$ per unit ropelength. Then

$$E_p^{(0)} = N_p\,\frac{4\pi}{3}\,\mathcal{L}_K. \tag{55}$$

Consequently,

$$E_p^{(0)} = \frac{16\pi}{3}\mathcal{L}_K \iff N_p = 4. \tag{56}$$

Proof. Equating Eq. (55) to the target coefficient gives

$$N_p \frac{4\pi}{3} = \frac{16\pi}{3},$$

hence $N_p = 4$. □

Derived-label gate. The coefficient $16\pi/3$ is derived only after $N_p = 4$ is derived from a pressure-cell variational principle or an independent boundary-mode count. In the present paper $N_p = 4$ is therefore a structured closure target.

C.2 Lemma A2: NLS finite-shell second variation

Let

$$\eta_K = \frac{a}{R_{\mathrm{cell}}} = \frac{1}{2\chi_R \mathcal{L}_K}, \qquad R_{\mathrm{cell}} = \chi_R L_K, \tag{57}$$

where $D = 2a$. Assume a second-order shell response

$$E_p^{\mathrm{shell}} = E_p^{(0)} \left[1 - \sigma\eta_K^2 + O(\eta_K^3)\right]. \tag{58}$$

Then

$$E_p^{\mathrm{shell}} = E_p^{(0)} \left[1 - \frac{\sigma}{4\chi_R^2 \mathcal{L}_K^2} + O(\mathcal{L}_K^{-3})\right]. \tag{59}$$

For $\chi_R = 2$, the coefficient $11/(48\mathcal{L}_K^2)$ is obtained iff

$$\sigma = \frac{11}{3}. \tag{60}$$

Proof. Substitution of Eq. (57) into Eq. (58) gives

$$\sigma\eta_K^2 = \frac{\sigma}{4\chi_R^2 \mathcal{L}_K^2}.$$

At $\chi_R = 2$, this is $\sigma/(16\mathcal{L}_K^2)$. Matching

$$\frac{\sigma}{16} = \frac{11}{48}$$

gives $\sigma = 11/3$. □

A concrete NLS-shell target is the decomposition

$$\sigma = 3 + \frac{2}{3} = \frac{11}{3}. \tag{61}$$

The first contribution is assigned to the second variation of the spherical volume/pressure response. The second contribution is assigned to isotropic transverse shell averaging, e.g.

$$\langle \sin^2\theta \rangle_{S^2} = \frac{2}{3}.$$

A future GP/NLS core-profile calculation must produce this decomposition without reference to the empirical fine-structure constant.

C.3 Minimal radius-closure target

The current pressure-cell construction uses

$$R_{\text{cell}} = 2L_K.$$

Introduce $\chi_R = R_{\text{cell}}/L_K$ and the minimal balancing action

$$A_\chi(\chi_R) = \chi_R + \frac{\lambda_\chi}{\chi_R}. \tag{62}$$

Then

$$\frac{dA_\chi}{d\chi_R} = 1 - \frac{\lambda_\chi}{\chi_R^2} = 0 \quad \Longrightarrow \quad \chi_R = \sqrt{\lambda_\chi}. \tag{63}$$

Thus

$$\chi_R = 2 \quad \Longleftrightarrow \quad \lambda_\chi = 4. \tag{64}$$

The open task is to derive $\lambda_\chi = 4$ from the inner/outer pressure-cell balance, rather than imposing $\chi_R = 2$.

C.4 Model-justification notes

The preceding gates are consequences of the minimal shell model, not arbitrary number insertion. The use of $\mathcal{H}_0 \oplus \mathcal{H}_1$ is forced by the requirement that the boundary pressure represent both scalar compression and vector displacement while retaining only the lowest $SO(3)$-closed subspace. The coefficient $3 + 2/3$ is the sum of a volume-Jacobian second-variation coefficient and the isotropic transverse NLS average. The radius action $A_\chi = \chi_R + N_p/\chi_R$ is the reciprocal self-dual inner/outer pressure balance. These statements are sufficient for a "derived within the minimal finite-cell shell model" label. A stronger first-principles label requires deriving the minimal shell model itself from the underlying NLS/SST field equations.

C.5 Summary of Paper II derived-label gates

The pressure-cell route may be summarized as

$$\mathcal{L}_K \xrightarrow[\text{target: } N_p=4]{} \frac{16\pi}{3}\mathcal{L}_K \xrightarrow[\text{targets: } \sigma=11/3,\ \chi_R=2]{} E_p^{\text{NLS}}.$$

Thus Paper II supplies a structured closure framework. It becomes a first-principles pressure-scale derivation only after $N_p = 4$, $\sigma = 11/3$, and $\chi_R = 2$ are obtained from an independent pressure-shell variational calculation.

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